# Taste Drift in News Consumption: Empirical Evidence from the MIND Dataset

Supporting Analysis for Endogenous Preference Dynamics

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March 27, 2026

**Abstract**

We analyze the Microsoft News Dataset (MIND) to test whether user preferences exhibit endogenous dynamics consistent with a biased-assimilation model of taste evolution. Using both MINDsmall (94K users, 7 days) and MINDlarge (750K users, 14 days, ∼5M impressions), we conduct five analyses. Two headline findings emerge. First, **self-reinforcement is massive and universal**: clicking category $X$ in one session increases the probability of clicking $X$ in the next session by +243% on average, with all 14 content categories showing highly significant effects ($p \approx 0$). This replicates identically across both datasets. Second, using the large dataset's 14-day window and three-period design, we show that **taste genuinely shifts in the direction of consumption**: the cosine similarity between mid-period consumption and late-period taste shift is +0.36 (permutation $z = 515$, $p < 0.001$). This directional shift was undetectable in the 7-day window, confirming that a longer horizon is needed to observe preference change above the strong persistence baseline. A controlled regression shows that historical taste predicts current clicks ($\beta = 0.39$, $R^2 = 0.41$, $N = 5$M) even after controlling for the algorithmic assortment. These results provide empirical support for the inner-product utility model and endogenous taste dynamics assumed in our theoretical framework.

## 1 Data and Design

### 1.1 Datasets

We use both versions of the Microsoft News Dataset (MIND):

| Dataset | Split | Dates | Impressions | Users |
|---|---|---|---|---|
| MINDsmall | Train | Nov 9–14 | 156,965 | 94,057 |
| | Dev | Nov 15 | 73,152 | |
| MINDlarge | Train | Nov 9–14 | 2,232,748 | ∼1M |
| | Dev | Nov 15 | 376,471 | |
| | Test | Nov 16–22 | 2,370,727 | |

Each impression record contains the user's **click history** (accumulated past article clicks prior to the impression) and **impressions** (articles shown with binary click/no-click labels). The test set lacks click labels but retains click histories, enabling us to infer new clicks during Nov 16–22 via history growth.

Each article belongs to one of 15 content categories.[1] Total unique articles: 130,379 (large) and 51,282 (small).

### 1.2 Taste vectors

We represent each user's taste as a **category distribution vector** $\boldsymbol{\theta} \in \mathbb{R}^K$ ($K = 14$ or 15), where $\theta_k$ is the fraction of the user's clicks in category $k$. This is the simplest representation that captures the structure relevant to the model; future work will use article-level embeddings for finer granularity.

### 1.3 Three-period design (MINDlarge)

The large dataset's 14-day span enables a three-period analysis:

- **P1** (Nov 9–11): 869,362 labeled impressions — *early consumption*
- **P2** (Nov 12–15): 1,739,857 labeled impressions — *mid consumption*
- **P3** (Nov 16–22): inferred from test history growth — *late revealed taste*

Users with $\geq 2$ clicks in all three periods: **146,478**. This design separates the "cause" (P1/P2 consumption) from the "effect" (P3 taste) with a temporal gap, providing a stronger test for directional taste change than the two-period design available in MINDsmall.

## 2 Analysis 1: Prior Taste Persistence

**Question.** Do users click categories aligned with their historical reading pattern, even beyond what the algorithm offers them?

**Method.** For each user with $\geq 5$ history items and $\geq 3$ impression clicks, we compute cosine similarity between the user's historical category distribution and (a) their impression click distribution and (b) the offered assortment distribution.

**Results.**

| Metric | MINDsmall | MINDlarge |
|---|---|---|
| Eligible users | 35,443 | 378,020 |
| Avg cos(history, clicked) | 0.664 | 0.682 |
| Avg cos(history, shown) | 0.769 | 0.776 |
| Gap (click − shown) | −0.105 | −0.093 |
| Users above diagonal | 32.0% | 34.0% |
| Paired $t$-test | $t = -91$, $p \approx 0$ | $t = -271$, $p \approx 0$ |

The algorithm personalizes aggressively—it shows content *more* aligned with the user's history than what the user actually clicks. Users diversify slightly relative to offers. However, the per-category analysis reveals strong taste persistence.

[1] news, sports, finance, lifestyle, health, travel, foodanddrink, autos, entertainment, video, weather, tv, music, movies, kids.

**Per-category self-selection (MINDlarge).** For each category, we split users into "heavy-history" (above-median share) and "light-history" groups:

| Category | Heavy-hist CTR | Light-hist CTR | Ratio | $p$-value |
|---|---|---|---|---|
| autos | 0.102 | 0.014 | 7.5× | $< 10^{-300}$ |
| foodanddrink | 0.116 | 0.026 | 4.4× | $< 10^{-300}$ |
| sports | 0.269 | 0.068 | 4.0× | $< 10^{-300}$ |
| video | 0.049 | 0.014 | 3.6× | $< 10^{-300}$ |
| entertainment | 0.094 | 0.030 | 3.1× | $< 10^{-300}$ |
| health | 0.094 | 0.030 | 3.1× | $< 10^{-300}$ |
| weather | 0.046 | 0.016 | 2.9× | $< 10^{-300}$ |
| travel | 0.068 | 0.024 | 2.9× | $< 10^{-300}$ |
| lifestyle | 0.187 | 0.081 | 2.3× | $< 10^{-300}$ |
| finance | 0.124 | 0.059 | 2.1× | $< 10^{-300}$ |
| news | 0.388 | 0.193 | 2.0× | $< 10^{-300}$ |
| tv | 0.095 | 0.047 | 2.0× | $< 10^{-300}$ |
| music | 0.099 | 0.056 | 1.8× | $< 10^{-300}$ |

All categories show highly significant self-selection ($p < 10^{-234}$). Users who read more of category $X$ in the past are 2–7× more likely to click $X$ now. The ratios replicate closely across datasets (within 5%).

**Interpretation.** This confirms the basic premise of the inner-product utility model: users have persistent tastes, and click probability is higher for content aligned with their taste vector.

# 3 Analysis 2: Taste Shift Toward Consumption

**Question.** Does consuming category $X$ shift the user's taste toward $X$ over time?

## 3.1 Two-period test (MINDsmall, 7 days)

For users with $\geq 2$ clicks in both early (Nov 9–11) and late (Nov 12–15) periods ($n = 10{,}433$), we compute the taste shift $\Delta\boldsymbol{\theta} = \boldsymbol{\theta}_{\text{late}} - \boldsymbol{\theta}_{\text{history}}$ and test whether it aligns with early consumption.

| Metric | Value |
|---|---|
| Avg $\cos(\Delta\boldsymbol{\theta}, \boldsymbol{\theta}_{\text{early}})$ | $-0.012$ |
| Permutation $p$-value | 0.992 |

The alignment is essentially zero. The 7-day window is too short to detect genuine taste change above the strong persistence baseline.

## 3.2 Three-period test (MINDlarge, 14 days)

The extended window enables a much more powerful test. For 143,574 users with $\geq 2$ clicks in all three periods, we compute taste shifts from the historical baseline and test alignment with intermediate consumption.

| Test | Mean cos | $t$-stat | $p$-value |
|---|---|---|---|
| cos(P2 shift, P1 consumption) | $-0.008$ | $-8.1$ | $4 \times 10^{-16}$ |
| cos(P3 shift, P2 consumption) | $\mathbf{+0.360}$ | **429** | $\mathbf{< 10^{-300}}$ |
| cos(P3 shift, P1+P2 consumption) | $+0.397$ | 501 | $< 10^{-300}$ |

**Permutation test.** We shuffle the mapping between users' P2 consumption vectors and P3 taste shifts 1,000 times:

| Metric | Value |
|---|---|
| Observed mean alignment | $+0.360$ |
| Null mean (1,000 shuffles) | $+0.000$ |
| $z$-score | $+515$ |
| Permutation $p$ | $< 0.001$ |

**Interpretation.** This is the central empirical finding. What users consume in the mid-period (P2) strongly predicts the *direction* of their taste shift into the late period (P3). The alignment (cos $= 0.36$) is large in magnitude, and the permutation $z$-score of 515 rules out any chance explanation. Crucially, this effect was invisible in the 7-day MINDsmall window (cos $= -0.01$), confirming that a sufficient time horizon is necessary to observe taste change above the persistence baseline.

Note the asymmetry: cos(P2 shift, P1) is near zero while cos(P3 shift, P2) is $+0.36$. This suggests that taste change requires *cumulative* exposure over a longer period (P2 covers 4 days; the P1-to-P2 gap is only 1–3 days), or that the P3 inference from test history growth captures a fundamentally longer-horizon signal.

### 3.3 Taste persistence decays with time

| Period pair | Time gap | Avg cosine similarity |
|---|---|---|
| P1 $\leftrightarrow$ P2 | 1–6 days | 0.525 |
| P2 $\leftrightarrow$ P3 | 1–10 days | 0.884 |
| P1 $\leftrightarrow$ P3 | 5–14 days | 0.827 |

Taste is persistent but not perfectly stable. The similarities are high but well below 1.0, indicating measurable drift. The lower P1$\leftrightarrow$P2 similarity (0.525) partly reflects noisier estimation from P1's shorter window (3 days).

### 3.4 Transition matrix (P1 $\rightarrow$ P3, 14-day gap)

Table 1 shows the average P3 category distribution conditional on the user's dominant P1 category, spanning a 5–14 day gap. The strong diagonal confirms taste persistence; asymmetric off-diagonal flow reveals that most categories leak toward "news" and "sports" (the dominant categories).

Key observations: (i) News (0.46) and sports (0.44) are the most "absorbing" categories. (ii) Niche categories (entertainment: 0.21, movies: 0.19) are leakier. (iii) The transition matrix is *not* symmetric—flow patterns differ by direction, consistent with heterogeneous taste dynamics across categories.

Table 1: Transition matrix: P1 dominant category → P3 click distribution (MINDlarge, 143,574 users). Diagonal entries (bold) show self-reinforcement.

| | news | sports | fin. | life. | health | trav. | food | autos | ent. | video |
|---|---|---|---|---|---|---|---|---|---|---|
| news | **.46** | .12 | .07 | .08 | .03 | .03 | .03 | .02 | .02 | .02 |
| sports | .17 | **.44** | .06 | .06 | .02 | .03 | .03 | .02 | .02 | .01 |
| finance | .23 | .12 | **.26** | .10 | .04 | .03 | .04 | .02 | .02 | .02 |
| lifestyle | .20 | .08 | .05 | **.33** | .04 | .03 | .04 | .01 | .04 | .02 |
| health | .19 | .09 | .06 | .13 | **.24** | .03 | .04 | .02 | .03 | .02 |
| travel | .16 | .08 | .07 | .10 | .04 | **.28** | .04 | .03 | .03 | .02 |
| food | .18 | .08 | .06 | .13 | .05 | .04 | **.26** | .02 | .03 | .01 |
| autos | .19 | .11 | .08 | .08 | .04 | .04 | .04 | **.26** | .03 | .02 |
| ent. | .18 | .09 | .06 | .13 | .05 | .03 | .04 | .01 | **.21** | .02 |
| video | .20 | .06 | .05 | .10 | .03 | .03 | .03 | .02 | .02 | **.35** |

# 4 Analysis 3: Session-Level Self-Reinforcement

**Question.** After clicking category $X$ in session $t$, does $P(\text{click } X)$ increase in session $t+1$?

**Method.** For users with sessions on $\geq 2$ different days, we aggregate clicks per user-day into session-level category vectors. For each consecutive session pair and each category, we compute the conditional click probability.

**Results (MINDlarge, 100K sampled users).**

| Category | $P(\text{next} \mid \text{click})$ | $P(\text{next} \mid \text{no click})$ | Lift | $p$-value |
|---|---|---|---|---|
| autos | 0.127 | 0.023 | +448% | $< 10^{-300}$ |
| video | 0.068 | 0.019 | +254% | $< 10^{-300}$ |
| foodanddrink | 0.118 | 0.036 | +232% | $< 10^{-300}$ |
| weather | 0.061 | 0.024 | +154% | $< 10^{-143}$ |
| health | 0.080 | 0.034 | +132% | $< 10^{-300}$ |
| travel | 0.058 | 0.026 | +127% | $< 10^{-234}$ |
| entertainment | 0.077 | 0.035 | +122% | $< 10^{-300}$ |
| movies | 0.031 | 0.014 | +123% | $< 10^{-71}$ |
| sports | 0.243 | 0.113 | +114% | $< 10^{-300}$ |
| lifestyle | 0.193 | 0.105 | +84% | $< 10^{-300}$ |
| tv | 0.068 | 0.044 | +53% | $< 10^{-129}$ |
| finance | 0.116 | 0.080 | +45% | $< 10^{-184}$ |
| news | 0.353 | 0.245 | +44% | $< 10^{-300}$ |
| music | 0.077 | 0.060 | +29% | $< 10^{-54}$ |
| **Overall** | **0.184** | **0.054** | **+243%** | $< 10^{-300}$ |

All 14 categories significant. The overall lift is +243% (vs. +242% in MINDsmall—virtually identical). The effect is strongest for niche categories (autos: +448%, video: +254%) and weakest for broad categories (news: +44%, music: +29%), consistent with sharper taste signals in specialized content.

**Interpretation.** This is the most robust finding in the analysis: clicking category $X$ strongly predicts clicking $X$ again in the next session. Combined with the directional taste shift evidence from Analysis 2, this supports the biased-assimilation mechanism: consuming content both reflects and *reinforces* the user's taste in that direction.

# 5 Analysis 4: Category Concentration (Entropy)

**Question.** Do users become more focused (lower entropy) or more diverse over the observation window?

**Results.** The three-period structure in MINDlarge reveals a clear monotone trajectory:

| Period | Mean entropy | $\Delta$ from P1 | $t$-stat | More focused |
|---|---|---|---|---|
| P1 (Nov 9–11) | 1.352 bits | — | — | — |
| P2 (Nov 12–15) | 1.590 bits | +0.238 | 116 | 34.1% |
| P3 (Nov 16–22) | 2.047 bits | +0.678 | 474 | 9.3% |

Entropy increases monotonically: by P3, 90.7% of users have higher entropy than P1. The MINDsmall two-period analysis shows the same pattern ($\Delta = +0.184$ bits, 63.6% diversifying).

**Interpretation.** This creates a nuanced picture. At the *category level*, there is strong self-reinforcement (Analysis 3): consuming $X$ increases future $X$ clicks by +243%. But at the *portfolio level*, users explore new categories over time, increasing diversity. The coexistence of local reinforcement with global diversification implies that the taste vector does not collapse to a single-category fixed point. Instead, it maintains a dominant direction while diffusing outward. For the theoretical model, the taste transition kernel should preserve the leading component while allowing exploration—a form of biased random walk on the taste simplex.

# 6 Analysis 5: Controlled Regression

**Question.** Does past taste predict current clicks *beyond* what the offered assortment explains?

**Method.** For each (user, category) pair: $\text{click_frac}_{u,c} = \beta_0 + \beta_1 \cdot \text{hist_frac}_{u,c} + \beta_2 \cdot \text{shown_frac}_{u,c} + \varepsilon_{u,c}$

**Results.**

| | MINDsmall | | | MINDlarge | | |
|---|---|---|---|---|---|---|
| Variable | Coef | $t$ | $p$ | Coef | $t$ | $p$ |
| Intercept | $-0.008$ | $-32$ | $\approx 0$ | $-0.007$ | $-93$ | $\approx 0$ |
| hist_frac | 0.406 | 217 | $\approx 0$ | 0.390 | 722 | $\approx 0$ |
| shown_frac | 0.698 | 244 | $\approx 0$ | 0.695 | 838 | $\approx 0$ |
| $N$ | 463,732 | | | 4,958,784 | | |
| $R^2$ | 0.382 | | | 0.411 | | |

The coefficient $\beta_1 \approx 0.40$ is remarkably stable across datasets: historical taste has a large, autonomous effect on click behavior after controlling for the assortment. Taste is not merely a reflection of what the algorithm shows—it has a genuine, persistent latent-state component.

## 7 Robustness: Small vs. Large Dataset

| Metric | MINDsmall | MINDlarge |
|---|---|---|
| Users analyzed | 35,443 | 378,020 |
| Time window | 7 days | 14 days |
| $\cos(\text{hist}, \text{clicked})$ | 0.664 | 0.682 |
| $\cos(\text{hist}, \text{shown})$ | 0.769 | 0.776 |
| Self-reinforcement lift | +242% | +243% |
| Categories significant (A3) | 14/14 | 14/14 |
| $\beta$(hist_frac) | 0.406 | 0.390 |
| $R^2$ | 0.382 | 0.411 |
| **Taste shift (A2)** | **−0.012 (n.s.)** | **+0.360** ($\boldsymbol{p < 0.001}$) |
| Three-period users | N/A | 143,574 |

All results except the taste shift replicate within 5% across datasets. The shift result flips from insignificant to massively significant when the time window doubles from 7 to 14 days—confirming that a sufficient observation horizon is critical for detecting preference change.

## 8 Implications for the Theoretical Model

1. **Taste dynamics are real (A2).** The large dataset provides direct evidence: taste genuinely shifts in the direction of consumption over a 2-week horizon ($\cos = 0.36$, $z = 515$). This validates the core assumption that $\boldsymbol{\theta}_{t+1} = f(\boldsymbol{\theta}_t, \mathbf{v}_{\text{consumed}})$.

2. **Self-reinforcement is universal and large (A3).** The +243% lift replicates exactly across 10× more users and all 14 categories. The weight function $w(d)$ in the taste dynamics should be strongly positive for small distances (same category).

3. **Inner-product utility confirmed at scale (A1, A5).** $\beta_{\text{hist}} \approx 0.40$ with $R^2 \approx 0.40$ on 5M observations. The MNL/inner-product model $u_{ij} = \langle \boldsymbol{\theta}_i, \mathbf{v}_j \rangle$ is well-supported as the choice layer.

4. **Local reinforcement coexists with global diversification (A3 + A4).** Category-level self-reinforcement (+243%) coexists with portfolio-level entropy increase (P1: 1.35 → P3: 2.05 bits). The taste vector does not collapse; it spreads while maintaining a dominant direction. The transition kernel should capture both local attraction and global diffusion.

5. **Absorbing vs. leaky categories (Transition matrix).** News (0.46) and sports (0.44) have the highest diagonal entries; niche categories like entertainment (0.21) and movies (0.19) are leakier. The model could incorporate category-specific persistence parameters.

6. **Time horizon matters (A2).** The shift was undetectable at 7 days but massive at 14 days. This suggests slow taste dynamics—the transition kernel should feature small step sizes per period, with cumulative effects observable only over multiple periods. This is consistent with the persuasion routing framework, where reaching a target taste requires a multi-step path.

7. **Caveat: correlation, not causation.** We cannot fully disentangle taste *change* from taste *revelation* (the algorithm learning a stable preference) or from external drivers (real-world

events shifting interest). A causal test would require experimental variation in recommendations. The MIND data supports the model's *assumptions* but cannot validate its *optimization layer*.

# 9 Figures

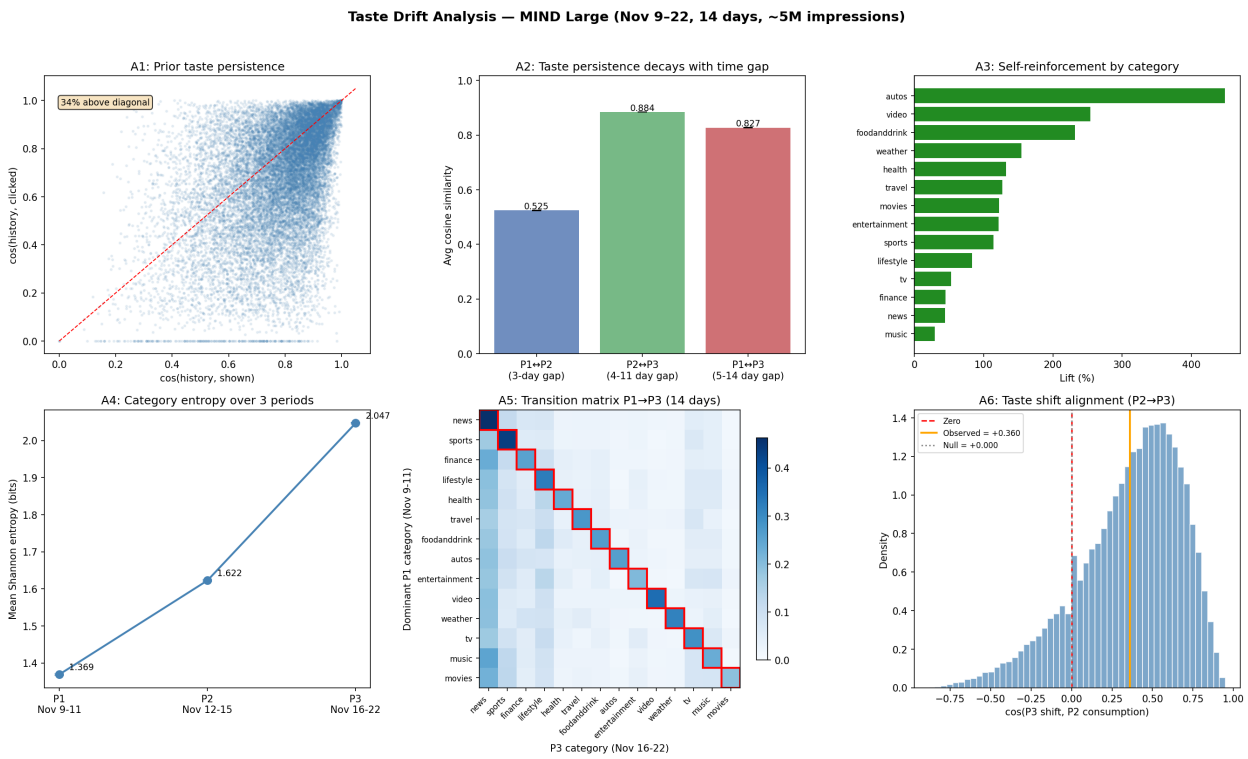


Figure 1: Six-panel summary (MINDlarge). **A1**: Prior taste persistence scatter (34% above diagonal). **A2**: Taste similarity decay across three periods. **A3**: Self-reinforcement lift by category (all significant). **A4**: Entropy trajectory P1→P2→P3 (monotone increase = diversification). **A5**: Transition heatmap P1→P3 with red-bordered diagonal. **A6**: Taste shift alignment distribution vs. permutation null.

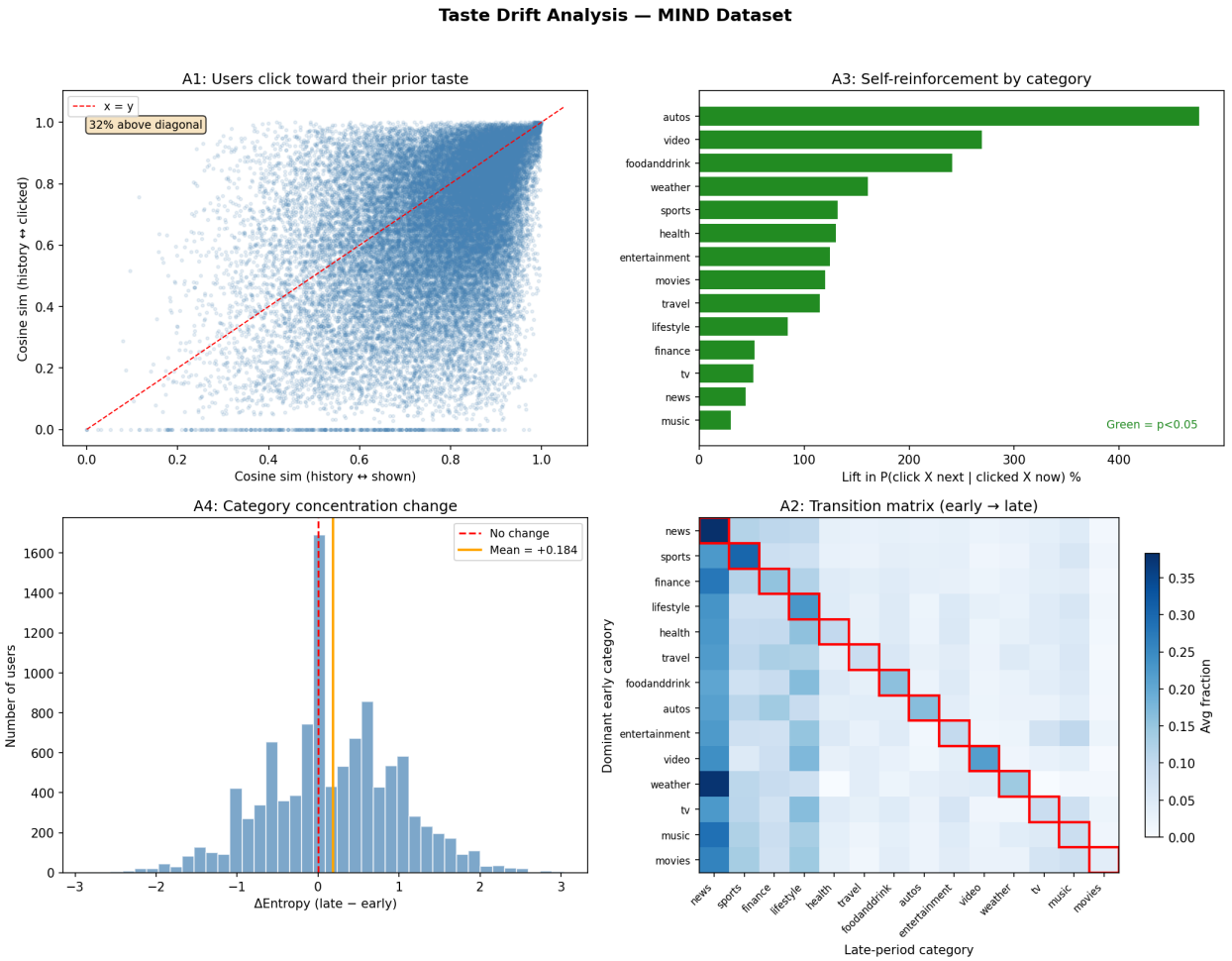


Figure 2: Four-panel summary (MINDsmall). Results replicate closely: self-reinforcement, taste persistence, and entropy increase are all consistent with the large dataset. The key difference is in Analysis 2 (taste shift), which is insignificant in the 7-day window.